

Chapter 10: Computer Graphics

**Computer Science: An Overview
Eleventh Edition**

**by
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Chapter 10: Computer Graphics

- 10.1 The Scope of Computer Graphics
- 10.2 Overview of 3D Graphics
- 10.3 Modeling
- 10.4 Rendering
- 10.5 Dealing with Global Lighting
- 10.6 Animation

2D Versus 3D Graphics

- **2D Graphics:** Deals with manipulating two-dimensional images
- **3D Graphics:** Deals with producing and displaying images of three-dimensional virtual scenes.

**Figure 10.1 A “photograph” of a virtual world
produced using 3D graphics** (from Toy Story by Walt Disney
Pictures/Pixar Animation Studios) © Corbis/Sygma

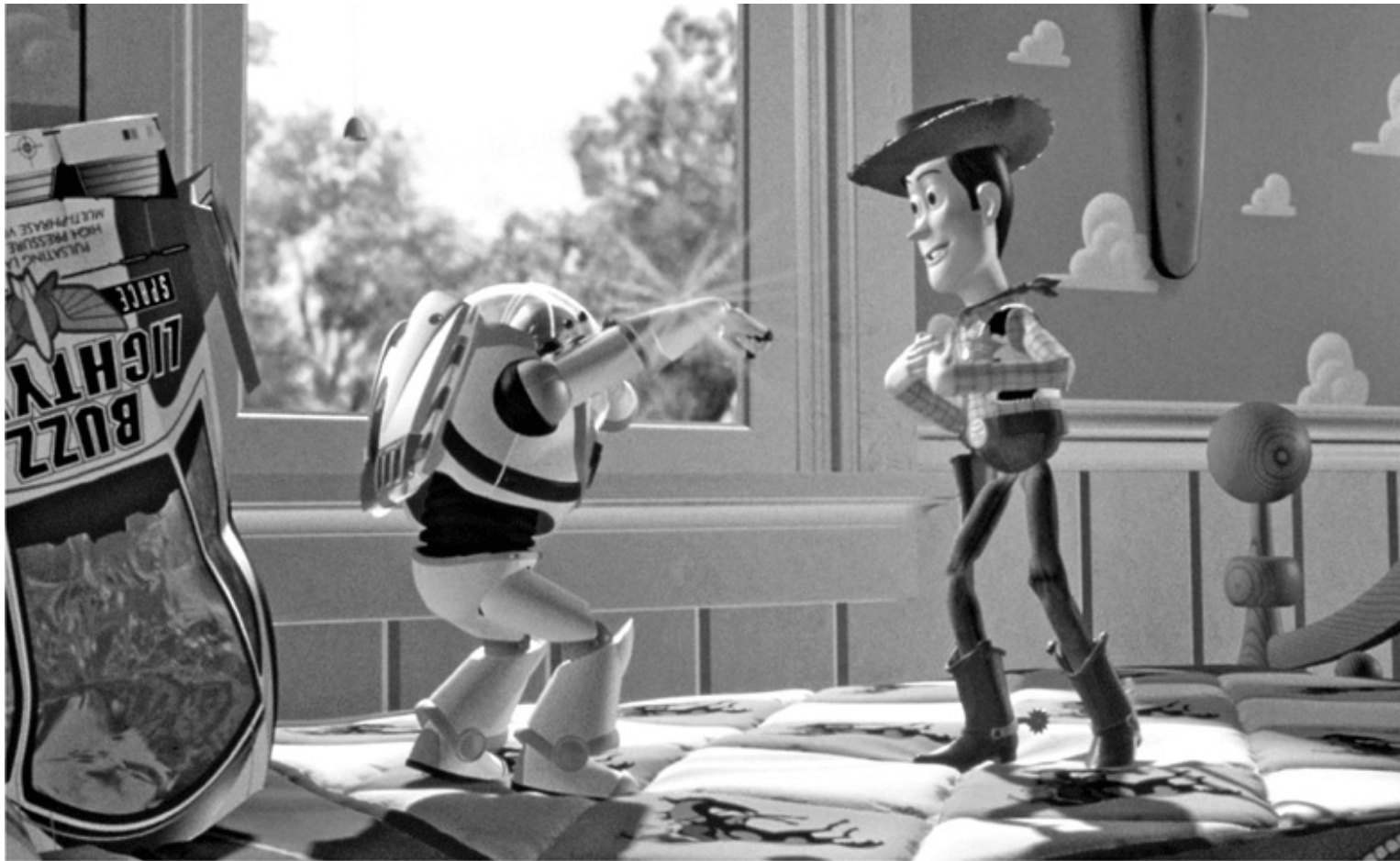
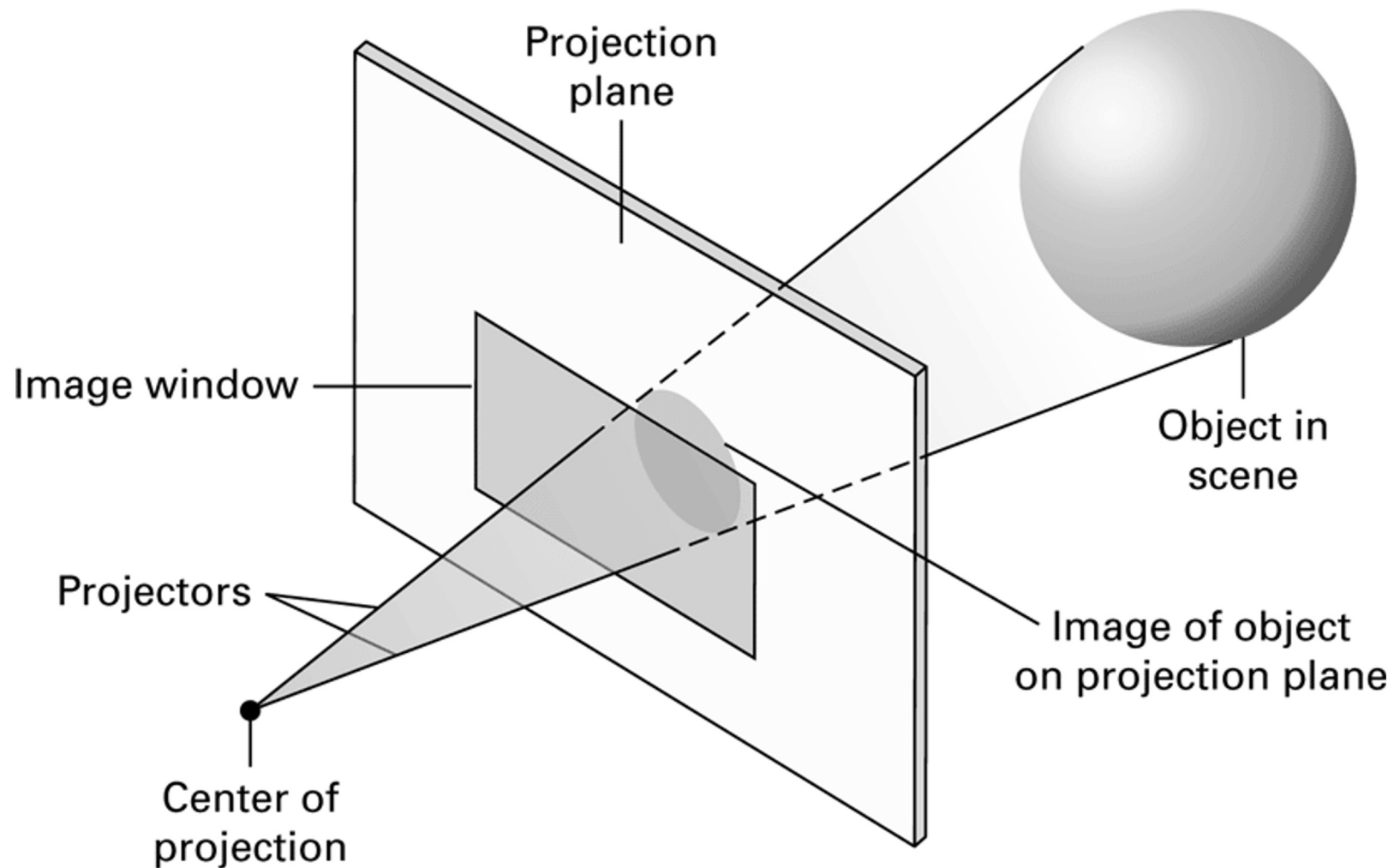


Figure 10.2 The 3D graphics paradigm



Modeling Objects

- Shape: Represented by a polygonal mesh obtained from
 - Traditional mathematical equations
 - Berzier curves and surfaces
 - Procedural models
 - Other methods being researched
- Surface: Can be represented by a texture map

Figure 10.3 A polygonal mesh for a sphere

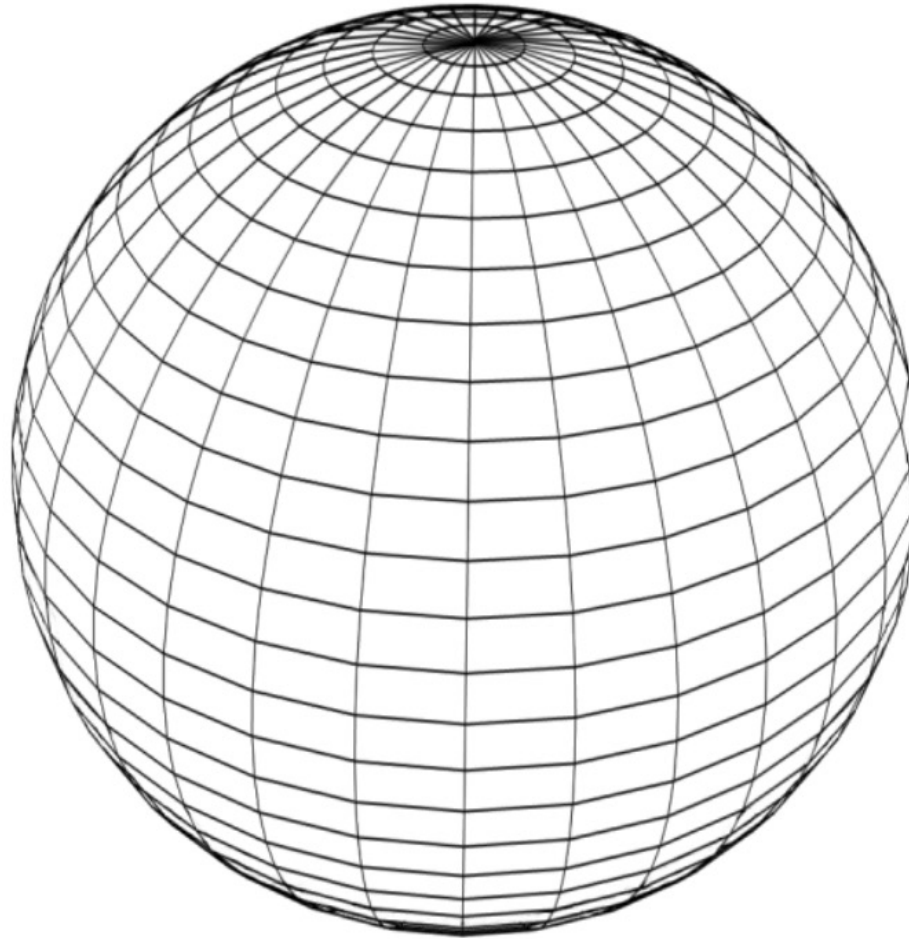


Figure 10.4 A Bezier curve

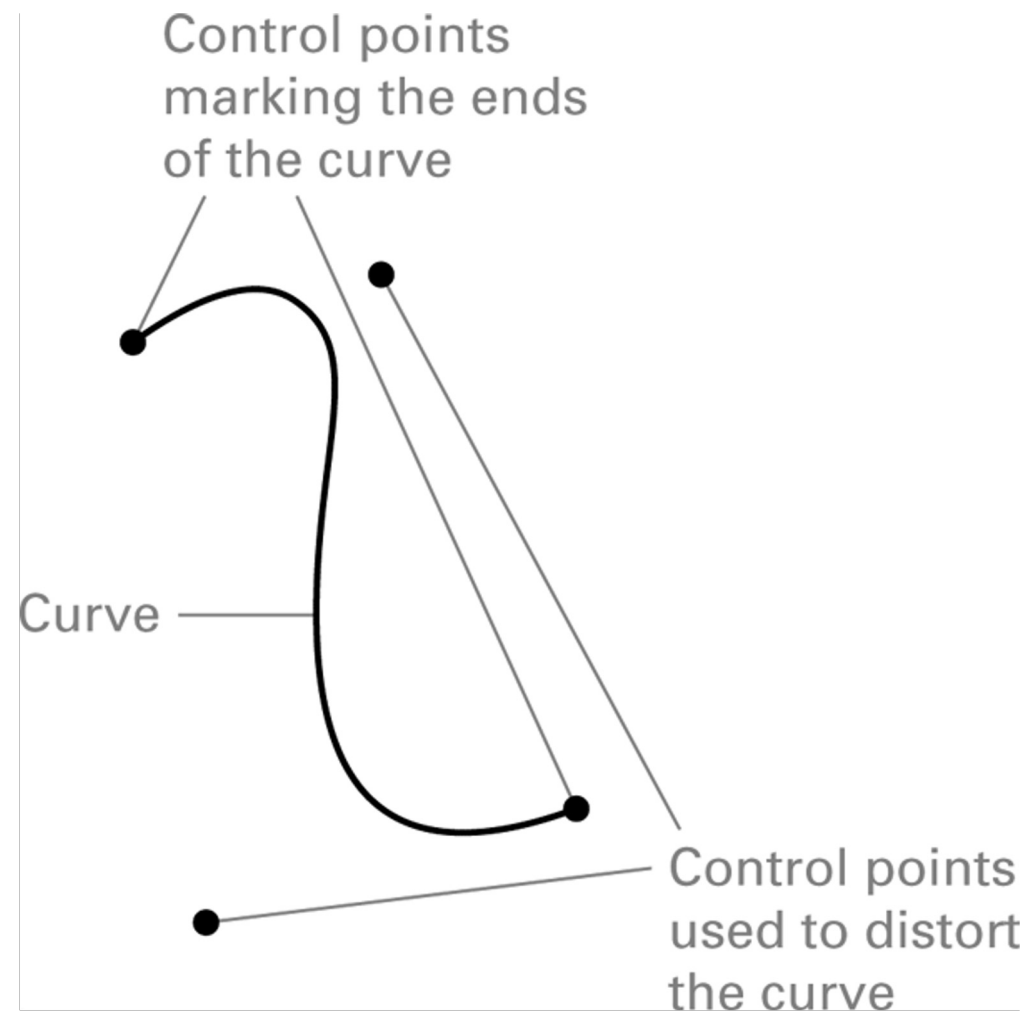
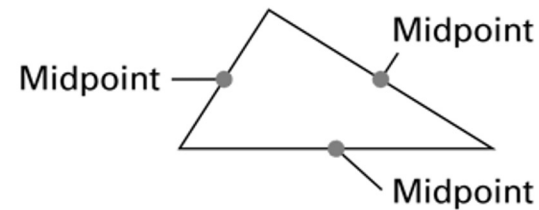


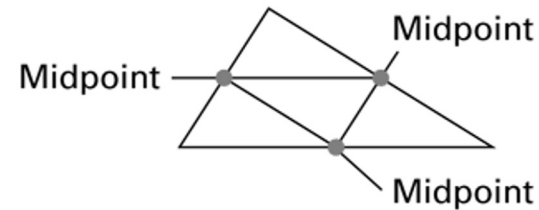
Figure 10.5

Growing a polygonal mesh for a mountain range

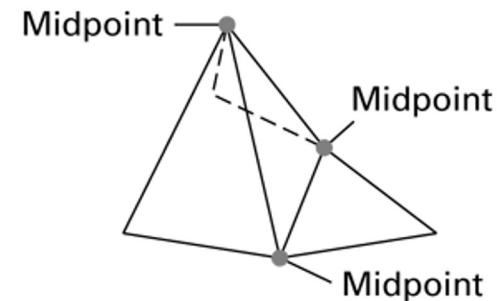
a. Identify the midpoints



b. Connect the midpoints



c. Move the midpoints



d. Repeat the process on the smaller triangles

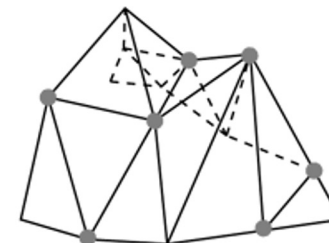


Figure 10.6 A scene from Shrek 2 by
Dreamworks SKG (© Dreamworks/The Kobal Collection)



Reflection Versus Refraction

- Reflection: Light rays bounce off surface.
 - Specular light
 - Diffuse light
 - Ambient light
- Refraction: Light rays penetrate surface.

Figure 10.7 Reflected light

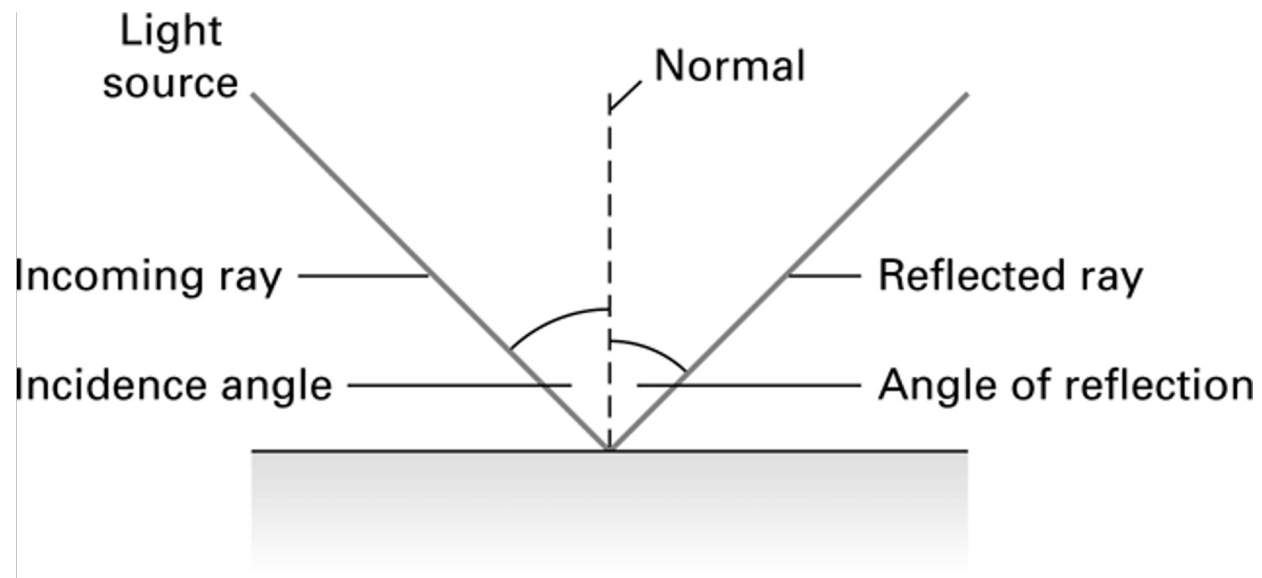


Figure 10.8 Specular versus diffuse light

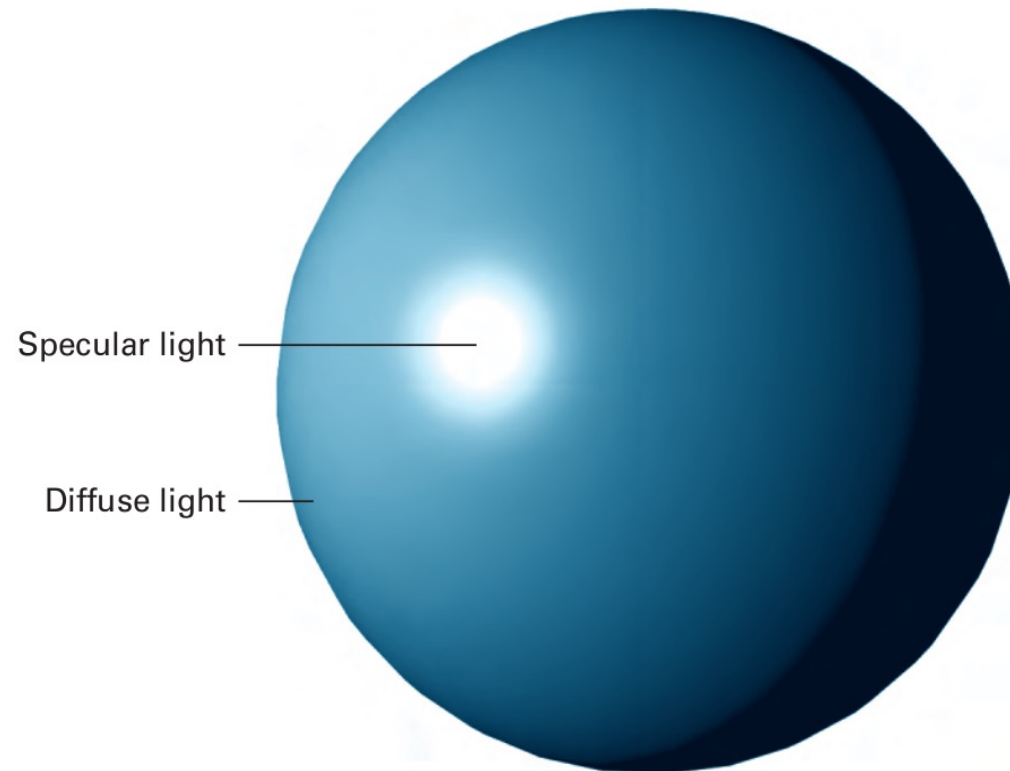
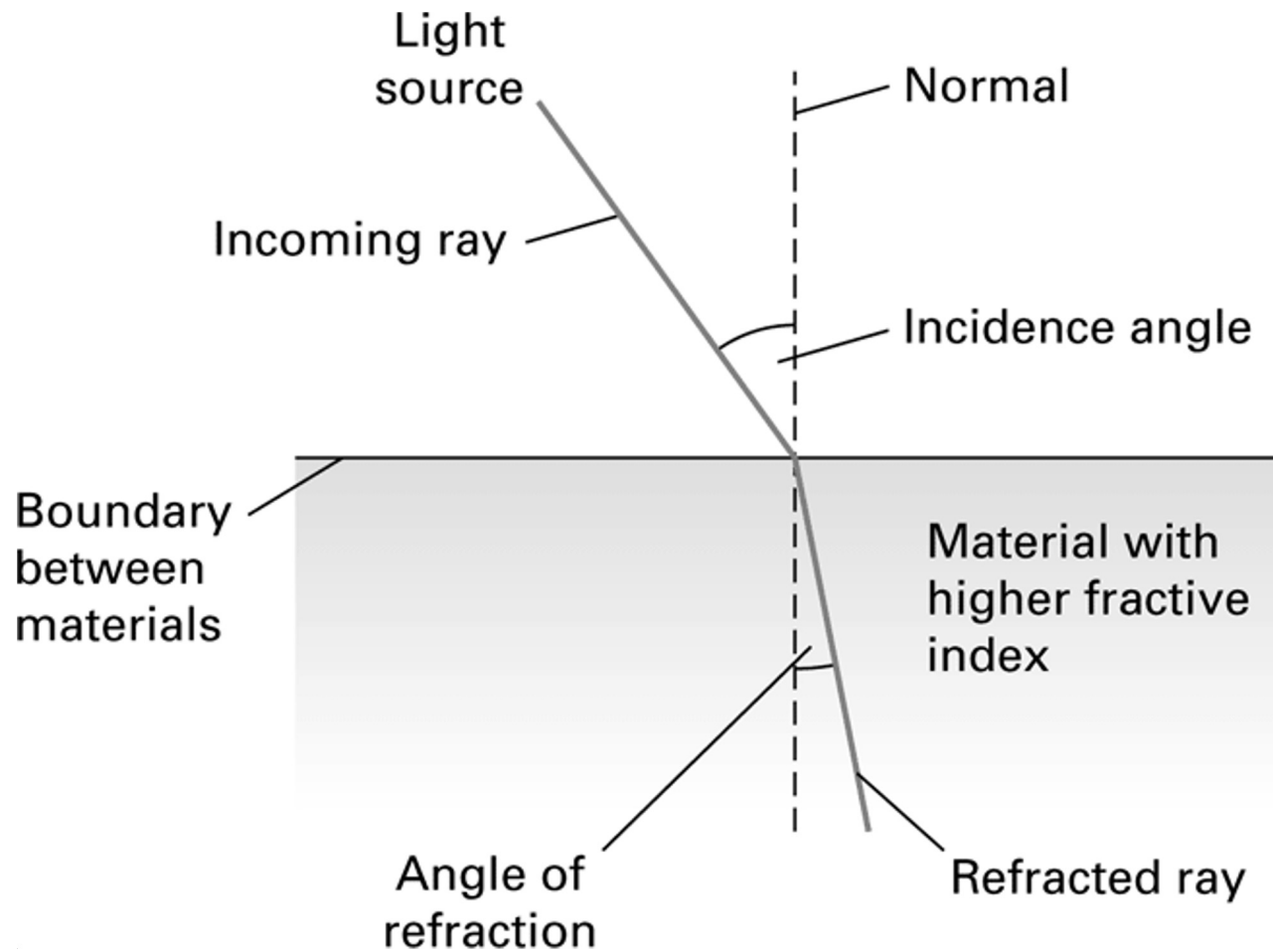


Figure 10.9 Refracted light



Rendering

- **Clipping:** Restricts attention to objects within view volume
- **Scan Conversion:** Associates pixel positions with points in scene
- **Shading:** Determines appearance of points associated with pixels

Figure 10.10 Identifying the region of the scene that lies inside the view volume

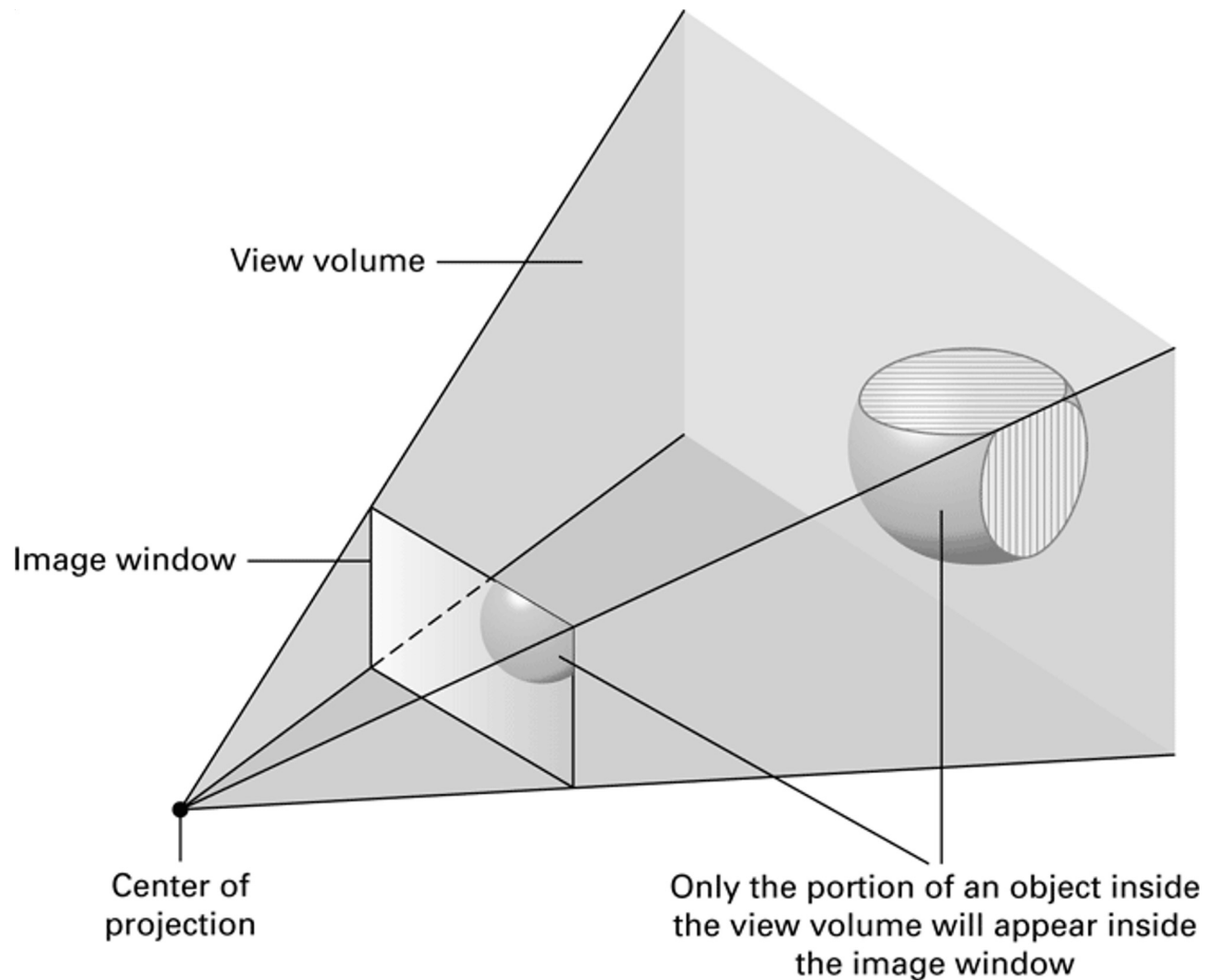
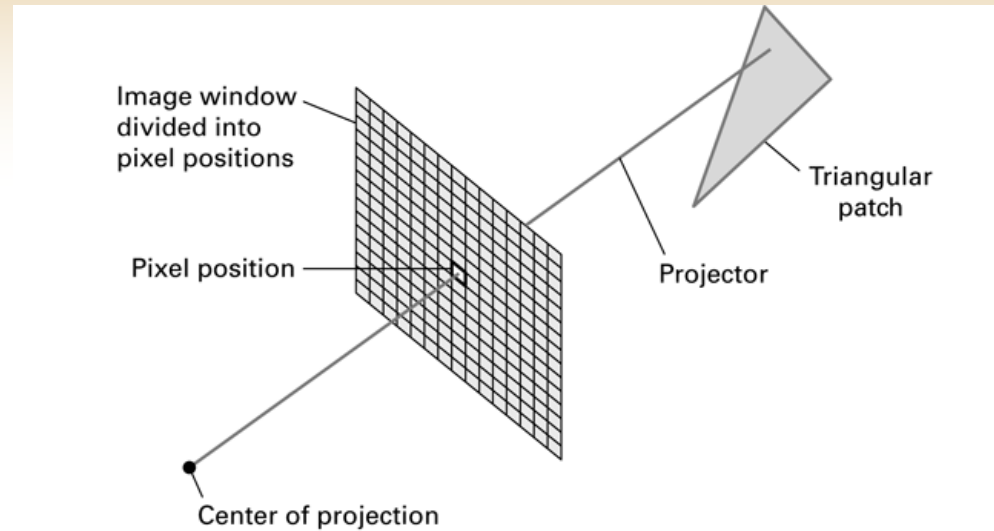
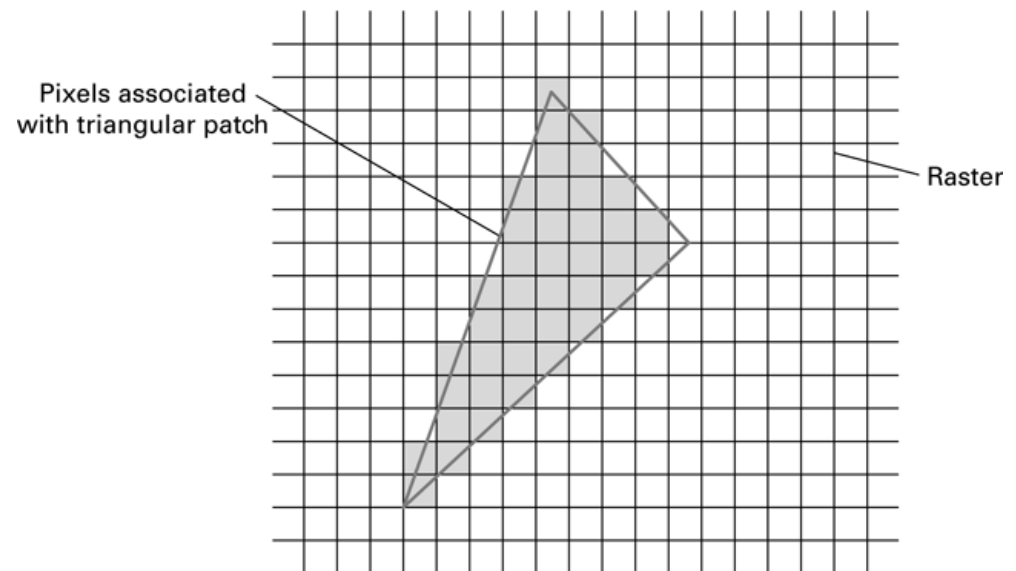


Figure 10.11

The scan conversion of a triangular patch



a. The scan conversion process



b. Raster showing the "projected shape" of the triangular patch

Shading Techniques

- **Flat Shading:** Creates faceted appearance
- **Gouraud and Phong Shading:** Creates smooth, rounded appearance
- **Bump Mapping:** Creates bumpy, rounded appearance

Figure 10.12 A sphere as it might appear when rendered by flat shading

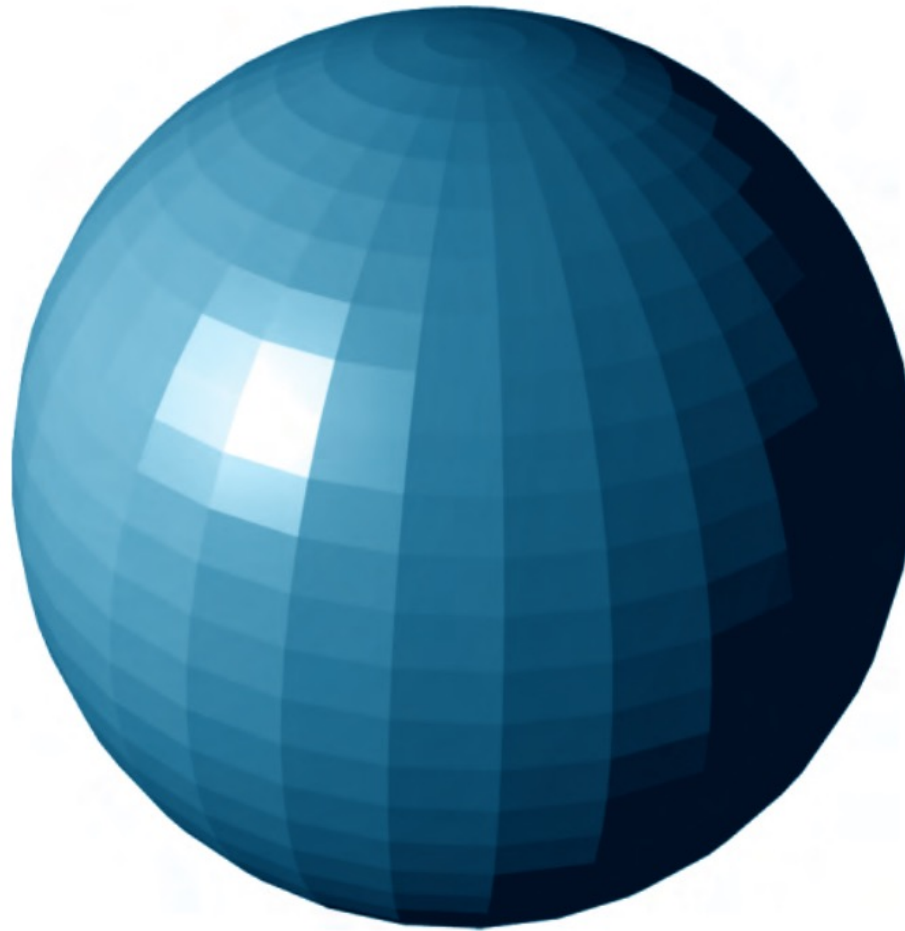


Figure 10.13 A conceptual view of a polygonal mesh with normal vectors at its vertices

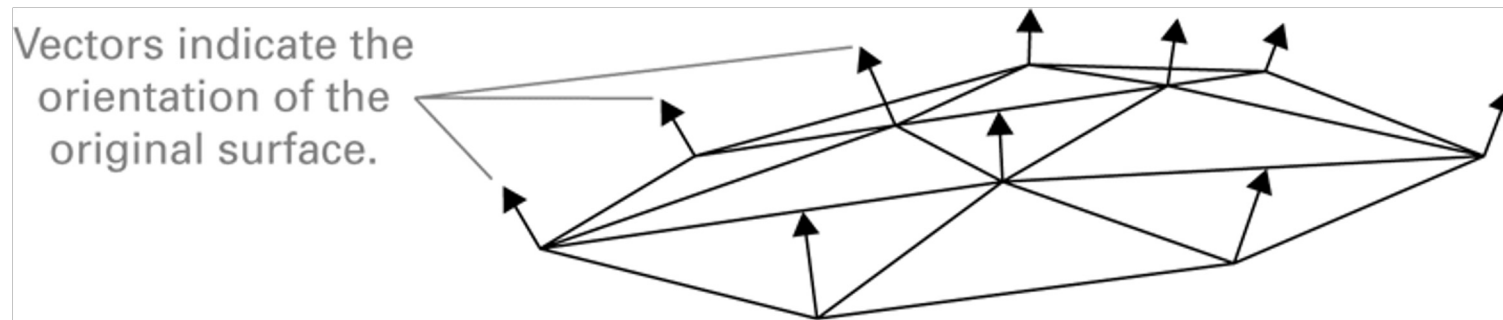
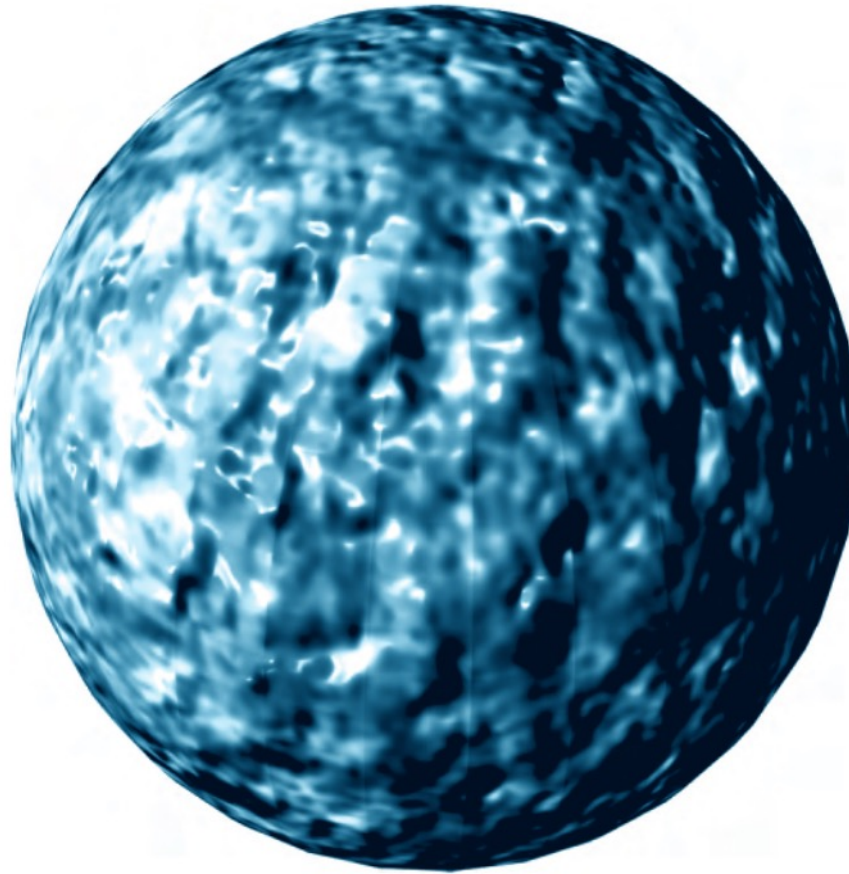


Figure 10.14 A sphere as it might appear when rendered using bump mapping



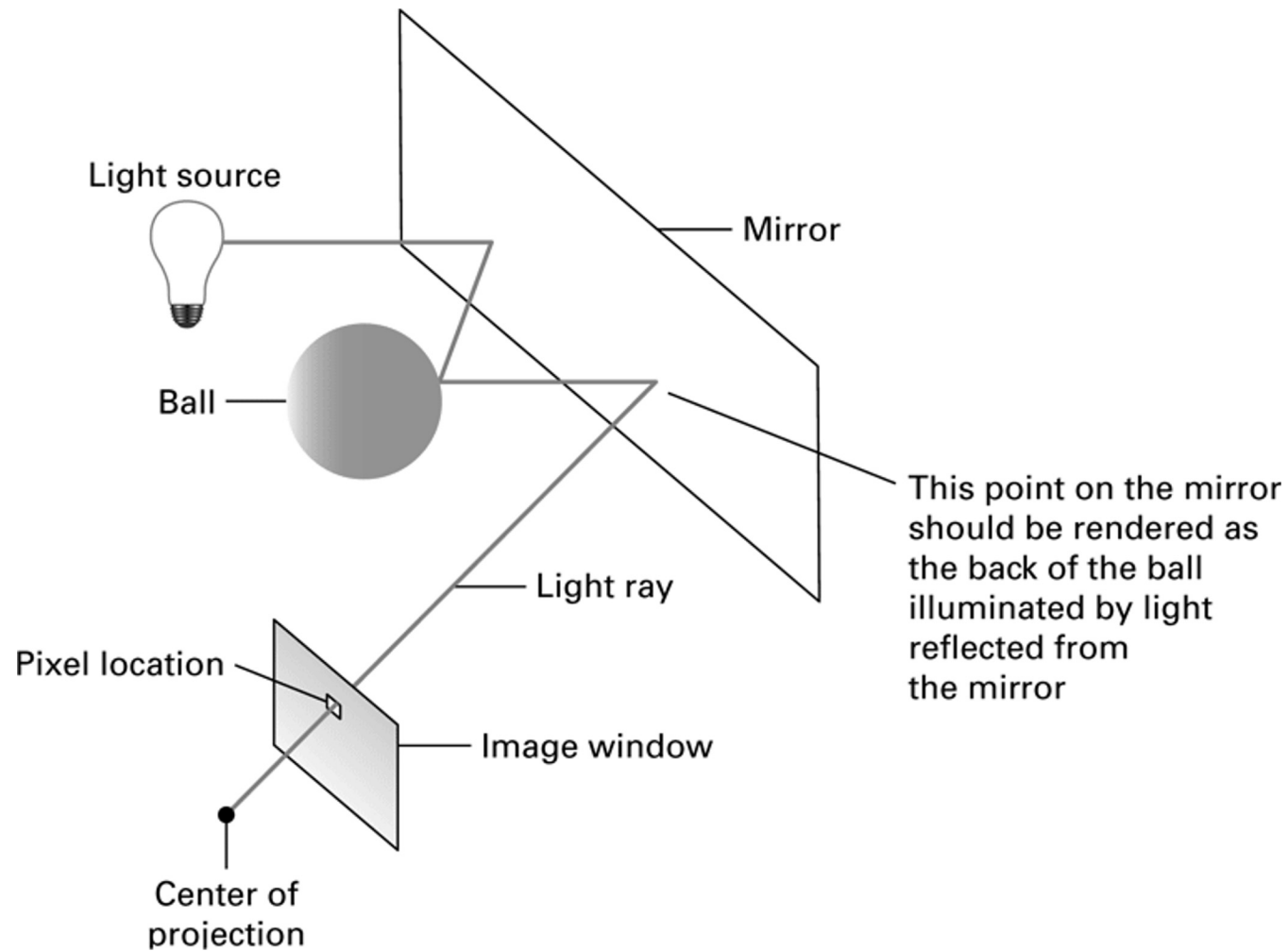
Rendering Pipeline

- Consists of traditional algorithms for clipping, scan conversion, and shading
- Often implemented in firmware
- Used as an abstract tool in graphics applications

Local Versus Global Lighting

- Local Lighting Model: Does not account for light interactions among objects
- Global Lighting Model: Accounts for light interactions among objects
 - Ray Tracing
 - Radiosity

Figure 10.15 Ray tracing



Animation

- **Storyboard:** A sequence of sketches summarizing the entire animation
- **Frame:** One of many images used to create animation
- **Key Frames:** Frames capturing the scene at specified points in time
- **In-betweening:** Producing frames to fill the gaps between key frames

Simulating Motion

- **Dynamics:** Applies laws of physics to determine position of objects
- **Kinematics:** Applies characteristics of joints and appendages to determine position of objects
 - Avars
 - Motion Capture